

Recommendation Systems for Personalized Content Discovery

Netflix Recommendation System using Collaborative Filtering and Matrix Factorization

Problem Statement

The objective is to build a personalized recommendation system capable of:

- Learning user preferences
- Predicting unseen movie ratings
- Generating Top-K recommendations
- Improving content discovery through personalization

Business Motivation

Recommendation systems power platforms such as Netflix, Spotify, Amazon, and YouTube by improving user engagement and retention.

This project explores how machine learning can be used to deliver personalized movie recommendations using the Netflix Prize Dataset.

Dataset Overview

Metric	Original Dataset
Ratings	100,480,507
Users	480,189
Movies	17,770
Rating Scale	1–5

Filtered Dataset Used

Metric	Value
Ratings	18,687,764
Users	148,083
Movies	4,491

Preprocessing

- Parsed Netflix Prize files
- Memory optimization using efficient data types
- Movie metadata integration
- User and movie filtering
- Date conversion and cleaning

Key Challenge

Dataset Sparsity = 97.19%

Density = 2.81%

Meaning most users interact with only a tiny fraction of available movies.

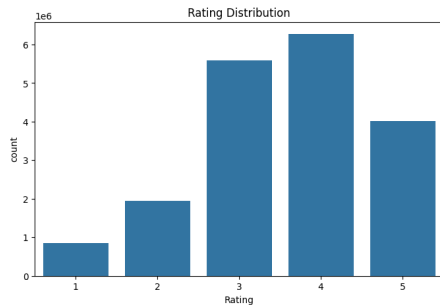


Figure: Rating Distribution

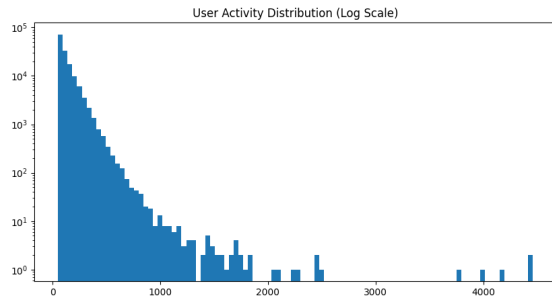


Figure: Distribution of User Activity

Key Insights

- Ratings are positively skewed toward 3–5 stars.
- User activity follows a long-tail distribution.
- A small number of highly active users contribute a large proportion of ratings.
- Dataset sparsity (97.19%) creates significant challenges for recommendation learning.

Model 1: Popularity-Based Recommendation

- Based on average rating and rating count
- Fast and simple
- No personalization

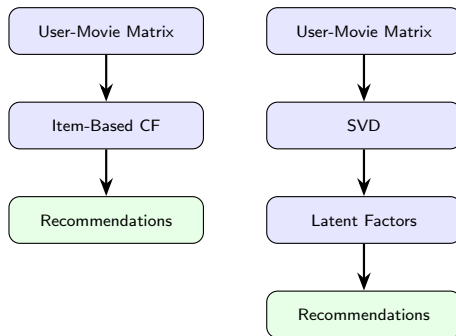
Model 2: Item-Based Collaborative Filtering

- Computes similarity between movies
- Generates personalized recommendations
- Sensitive to sparsity

Model 3: Matrix Factorization (SVD)

- Learns latent user and movie factors
- Captures hidden preference patterns
- Better suited for sparse datasets

Workflow Diagram



Train-Test Strategy

- 80:20 Train-Test Split

Mandatory Metrics

- **RMSE**: Measures rating prediction accuracy
- **MAP@10**: Measures recommendation ranking quality. Relevant if rating ≥ 3.5

Additional Metric

- **Precision@10**: Measures recommendation relevance

Why Ranking Metrics Matter

A recommendation system should not only predict ratings accurately but also rank relevant movies at the top of recommendation lists.

Results Table

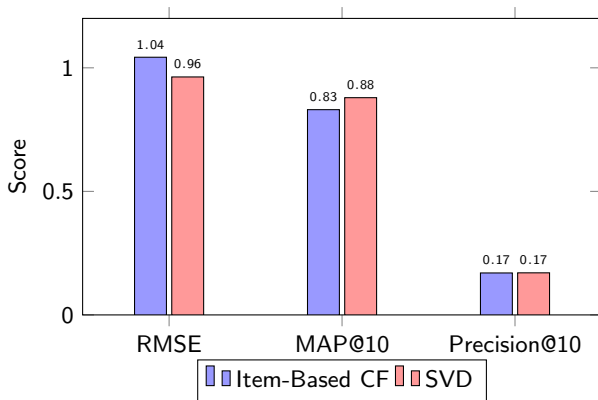
Model	RMSE	MAP@10	Precision@10
Item-Based CF	1.0429	0.8307	0.1698
SVD	0.9631	0.8792	0.1703

Conclusion

SVD consistently outperformed Item-Based Collaborative Filtering across prediction accuracy, ranking quality, and recommendation relevance. Best Model: SVD (RMSE = 0.9631, MAP@10 = 0.8792)

Model performance comparison

Comparison of recommendation models across evaluation metrics.



Selected User: User ID: 305344

User's Top Rated Movies

Movie	Rating
Yanni: Live at the Acropolis	5
18 Again	5
Yellow Submarine	5
Lord of the Rings: Fellowship of the Ring	5

Recommended Movies Generated by SVD

Movie	Predicted Rating
Doctor Zhivago	3.5082
Nuremberg	2.8792
Ray	2.8545
The Incredibles: Bonus Material	2.6354

Observation

The SVD model successfully identified unseen movies by leveraging latent user preferences rather than relying solely on popularity trends.

Key Insights

- Dataset sparsity is the primary challenge.
- Popularity-based methods lack personalization.
- Item-Based CF improves personalization but struggles with sparse data.
- SVD achieved the best performance across RMSE, MAP@10, and Precision@10.

Failure Cases

- New users (cold start)
- New movies with limited interactions
- Preference drift over time
- Sparse interaction histories

Future Improvements

- Neural Collaborative Filtering
- Hybrid Recommendation Systems
- Explainable Recommendations
- Real-Time Recommendation Pipelines

Final Takeaway

Matrix Factorization (SVD) proved to be the most effective recommendation approach, achieving the best balance between rating prediction accuracy and recommendation quality on the Netflix Prize Dataset.